Instrumental variables

ECON726

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Housekeeping

Agenda:

1. Discuss final project
2. Instrumental variables. Notes from:
   - *Business Data Science*, chapter 6
   - https://matheusfacure.github.io/python-causality-handbook/08-Instrumental-Variables.html
   - Notes from PhD course on linear regression (taught by Jim Hamilton)
   - Code available here: https://github.com/tara-sullivan/econ726-notebooks/blob/main/notebooks/11-iv.ipynb

Notes on emails:

- Email policy: I'll reply within two business days, and do my best to reply in one
- Please note that in general, I cannot reply to emails during business hours (8AM - 5PM on weekdays)

Next week: guest lecture on NYC open data

- I will take attendance!

Empirical project

Paper selection: Due **September 22** [5% of final grade]

- Done!

Project plan: Due **October 14** [15% of final grade]

- Done!

Replication results: **November 7**: [10% of final grade]

- Done!

Presentation: Due **December 2** & **December 9** [10% of final grade]

- Prepare a 5-8 minute presentation on your paper. Focus on the research question, main result, and one extension.

Final Paper: Due **December 16** [35% of final grade]

- Your paper should contain 6-10 pages of text, and 3-6 pages of figures (double spaced, 12 point font). You must provide the code and data to replicate your findings. Your final project should be submitted on Brightspace, and your code and data should be uploaded to github (note: if your data is too large to upload to github, let me know).

Replication results

- I have not had time to grade your replication results yet. I'm hoping to complete that this weekend, though it may take longer
- In general, this will be graded more easily, and the code in your final paper will be graded strictly

Presentation rubric (preliminary rubric)

10% of final grade; 1% for time limit

1. 1%: stay within time limit
2. 3%: Present research question clearly
3. 3%: Present key results clearly
4. 3%: Present extension, or present a substantive critique of paper, focusing on causal analysis; is this paper causal?

Helpful context:

- Pretend you were the author and are giving the elevator pitch of this paper
- You essentially have time for 3 slides (maybe 4 if you're stretching it!)

Final paper (preliminary rubric)

35% of final grade

Overview of paper: (6%)

- Summarize the public policy or business context of your replication paper. Why are the results valuable?
- Discuss the data you are using in detail. Be specific, and focus on your primary result. What are the raw data sources? What data sets are being merged together? What are the limitations of this data?
- Discuss the primary finding of this paper that you plan on. Be specific (for example, identify what tables or figures are the primary findings)
- When you describe results, you should use both technical language and prose (the significant coefficient X implies a X% change in outcome)

Final paper rubric (preliminary rubric)

Causality: provide a formal discussion of causality (9%)

- What is the ideal randomized experiment
- What is the method used
- Why this method is causal
  - The authors will say "I use differences-in-differences to identify..." You cannot get away with that. You need to motivate why your method is causal. If you don't discuss why your method is causal, you will lose all of these points.
- Include citations to materials from the course. You will likely need to cite alternative materials as well
- Penalty: reference methodologies you don't understand or motivate why they matter

Final paper rubric (preliminary rubric)

Code: 8%

1. 5%: Can I run your code? **SHARE YOUR CODE WITH CLASSMATES TO CHECK THIS**
2. 2%: Quality of code. Did you distill code to key results? Is your code readable?
3. This will be graded strictly! If I can't run your code with minimal edits, you don't get credit

Extension: 12%

1. Complete some sort of extension. This should be ambitious. This is your opportunity to differentiate yourself in this class!

Outline

Endogeneity problem

Simultaneous Equation Bias

Instrumental variables

IV estimation

Endogeneity problem

Simultaneous Equation Bias

Instrumental variables

IV estimation

Instrumental variables

In big complicated systems, there are many inputs that determine outputs

- Understanding causation in such systems requires **randomization**

$\Longrightarrow$ We want events that **look** like experiments

Common situation: the treatment has not been directly randomized, but rather **indirectly randomized** through experiments on variables that influence policy selection

- Indirect randomizers are **instrumental variables**

Example: In a drug trial, participants are randomly assigned to be given some drug

- Problem: Not every treated patient will take the drug
  - Drug may be painful or inconvenient, or people forget
- Treatment may be correlated with unobserved factors that influence outcome
  - Sicker patients may find the drug more painful, and be less likely to take the drug
  - In this case, you can't easily control for relevant differences between drug-takers and non-takers

"*Intent-to-treat*" problem

IV as solution to intent-to-treat problem

Solution to intent-to-treat problem:

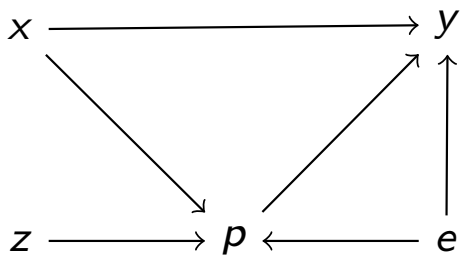
- Randomization changes the **probability** that a user takes a drug
  - If participant is not assigned drug access, there's a small or zero probability they take the drug
  - If participant is assigned drug access, there's a larger probability they take the drug

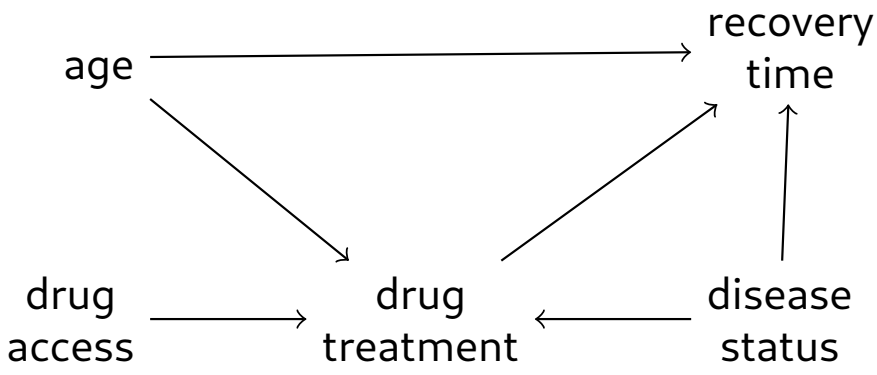
$\Longrightarrow$ IV models how change in probabilities affects outcomes

Overview of problem

Set-up:

- y: response of interest
- p: treatment variable
- x: observable covariates that directly influence y
- z: instrument; only affects response through influence on treatment
- e: unobserved factors that influence treatment and response





Key takeaways from model:

- Instrument z only impacts outcome y through treatment p
- Instrument z is completely independent of unobserved error e

IV regression as a structural model

Write causal model as a regression:

$$y = g(x, p) + e, \quad \text{Cov}(e, p) \neq 0$$

where:

- $g(x, p)$: structural function; represents how p acts causally on y
- $\text{Cov}(e, p) \neq 0$: error and policy are correlated
  - This is different than traditional regression model, where $\mathbf{E}[e|p] = 0$
  - Here, unobserved error influences treatment, so you don't have conditional independence

$\text{Cov}(e, p) \neq 0$ is different than traditional regression model

- If $\text{Cov}(e, p) = 0$, policy is **exogenous**
- In this set-up, policy is **endogenous**

Regression with endogenous covariates

Suppose you run a regression, when the true structural form is:

$$y = g(x, p) + e, \quad \text{Cov}(e, p) \neq 0$$

The [non-causal] conditional expectation would be:

$$\begin{aligned}\mathbf{E}[y|x, p] =& \mathbf{E}[g(x, p) + e|x, p] \\ =& \mathbf{E}[g(x, p)|x, p] + \mathbf{E}[e|x, p] \\ =& g(x, p) + \mathbf{E}[e|p]\end{aligned}$$

Regression would uncover:

- True structural relationship, plus
- Omitted variable bias

Next, we'll consider an alternative source of endogeneity: simultaneous equation bias

Endogeneity problem

Simultaneous Equation Bias

Instrumental variables

IV estimation

Example: supply and demand

Define Q_t as the quantity or oranges sold at time t, and P_t be the price of oranges sold at time t

- **Goal:** estimate the demand curve to determine how a higher price affects quantity demanded by consumers:

$$Q_t = \alpha + \beta P_t + \epsilon_t^d \tag{1}$$

To make math easier, first write (1) at the mean:

$$\bar{Q} = \alpha + \beta \bar{P} + \bar{\epsilon}$$

Demeaning Q_t (subtracting the mean) removes the intercept. Let the lowercase letter be the de-meaned variable:

$$q_t = \beta p_t + \epsilon_t^d$$

Example: supply curve

Consider analogous supply curve:

$$q_t = \gamma p_t + \epsilon_t^s$$

In this example:

- γ isolates the effect of a price change on how many oranges are brought to the market
- Variations in ϵ_t^s comes from things like weather

Under classical regression assumptions:

$$\mathbf{E}[\epsilon_t^d] = 0 \qquad \mathbf{E}[\epsilon_t^s] = 0$$
$$\mathbf{E}[\epsilon_t^d]^2 = \sigma_d^2 \qquad \mathbf{E}[\epsilon_t^s]^2 = \sigma_s^2$$
$$\mathbf{E}[\epsilon_t^d \epsilon_t^s] = 0$$

Combine demand and supply

Demand and supply curve **system of equations**:

$$\begin{aligned} \textbf{Demand:} \quad q_t &= \beta p_t + \epsilon_t^d, \\ \textbf{Supply:} \quad q_t &= \gamma p_t + \epsilon_t^s \end{aligned}$$

Setting them equal to each other:

$$\begin{aligned} \gamma p_t + \epsilon_t^s &= \beta p_t + \epsilon_t^d \\ \gamma p_t - \beta p_t &= \epsilon_t^d - \epsilon_t^s \\ p_t(\gamma - \beta) &= \epsilon_t^d - \epsilon_t^s \\ \implies p_t &= \frac{\epsilon_t^d - \epsilon_t^s}{\gamma - \beta} \end{aligned}$$

Consider dynamics associated with this equation:

- Suppose screwdrivers become a popular drink. This would be a shock to demand, and cause prices to rise:

$$\epsilon_t^d \uparrow \implies p_t \uparrow$$

- Consider if the weather is particularly good for growing oranges one season:

$$\epsilon_t^s \uparrow \implies p_t \downarrow$$

Dynamics associated with quantity demanded

Demand curve:

$$q_t = \beta p_t + \epsilon_t^d \tag{2}$$

We've shown:

$$p_t = \frac{\epsilon_t^d - \epsilon_t^s}{\gamma - \beta} \tag{3}$$

Substituting (3) into (2):

$$\begin{aligned} q_t &= \beta p_t + \epsilon_t^d \\ &= \beta \left(\frac{\epsilon_t^d - \epsilon_t^s}{\gamma - \beta} \right) + \epsilon_t^d \\ &= \frac{\beta}{\gamma - \beta} \epsilon_t^d - \frac{\beta}{\gamma - \beta} \epsilon_t^s + \epsilon_t^d \\ &= \left(\frac{\gamma}{\gamma - \beta} \right) \epsilon_t^d - \left(\frac{\beta}{\gamma - \beta} \right) \epsilon_t^s \end{aligned}$$

Consider dynamics associated with this equation:

- Suppose screwdrivers become a popular drink. This would be a shock to demand, and cause prices to rise:

$$\epsilon_t^d \uparrow \implies q_t \uparrow$$

- Consider if the weather is particularly good for growing oranges one season:

$$\epsilon_t^s \uparrow \implies q_t \downarrow$$

Regression coefficient

$$\text{plim of the OLS estimate of the price coefficient} = \frac{Cov(p_i, q_i)}{Var(p_i)}.$$

This implies:

$$\hat{\beta} = \frac{\sum q_t p_t}{\sum p_t^2} = \frac{T^{-1} \sum q_t p_t}{T^1 \sum p_t^2} \xrightarrow{p} \frac{\mathbf{E}(p_t q_t)}{\mathbf{E}(p_t^2)}.$$

Estimated coefficient: numerator

$$\hat{\beta} \to \frac{\mathbf{E}\left(p_t q_t\right)}{\mathbf{E}\left(p_t^2\right)}$$

Given our simplified demand and supply equations:

$$p_t = \frac{\epsilon_t^d - \epsilon_t^s}{\gamma - \beta}$$

$$q_t = \left(\frac{\gamma}{\gamma - \beta}\right)\epsilon_t^d - \left(\frac{\beta}{\gamma - \beta}\right)\epsilon_t^s$$

Consider the numerator and denominator separately. First the numerator:

$$\begin{aligned}
\mathbf{E}\left(p_t q_t\right) =& \mathbf{E}\left[\left(\left(\frac{1}{\gamma - \beta}\right)\epsilon_t^d - \left(\frac{1}{\gamma - \beta}\right)\epsilon_t^s\right)\left(\left(\frac{\gamma}{\gamma - \beta}\right)\epsilon_t^d - \left(\frac{\beta}{\gamma - \beta}\right)\epsilon_t^s\right)\right] \\
=& \mathbf{E}\left[\left(\frac{\gamma}{(\gamma - \beta)^2}\right)\left(\epsilon_t^d\right)^2 + \left(\frac{\beta}{(\gamma - \beta)^2}\right)(\epsilon_t^s)^2\right] \\
=& \frac{\gamma}{(\gamma - \beta)^2} \cdot \sigma_d^2 + \frac{\beta}{(\gamma - \beta)^2} \cdot \sigma_s^2
\end{aligned}$$

Estimated coefficient: denominator

$$\hat{\beta} \to \frac{\mathbf{E}\left(p_t q_t\right)}{\mathbf{E}\left(p_t^2\right)}$$

Given our simplified demand and supply equations:

$$p_t = \frac{\epsilon_t^d - \epsilon_t^s}{\gamma - \beta}$$

$$q_t = \left(\frac{\gamma}{\gamma - \beta}\right)\epsilon_t^d - \left(\frac{\beta}{\gamma - \beta}\right)\epsilon_t^s$$

Next consider the denominator:

$$\begin{aligned}
\mathbf{E}\left(p_t^2\right) =& \mathbf{E}\left[\left(\frac{\epsilon_t^d}{\gamma - \beta} - \frac{\epsilon_t^s}{\gamma - \beta}\right)^2\right] \\
=& \mathbf{E}\left[\frac{1}{\gamma - \beta} \cdot \left(\epsilon_t^d\right)^2 + \frac{1}{\gamma - \beta}\left(\epsilon_t^s\right)^2\right] \\
=& \frac{1}{(\gamma - \beta)^2} \cdot \sigma_d^2 + \frac{1}{(\gamma - \beta)^2}\sigma_s^2
\end{aligned}$$

Estimated coefficient

$$\hat{\beta} \to \frac{\mathbf{E}\left(p_t q_t\right)}{\mathbf{E}\left(p_t^2\right)}$$

We've shown:

$$\mathbf{E}\left(p_t q_t\right) = \frac{\gamma}{(\gamma-\beta)^2} \cdot \sigma_d^2 + \frac{\beta}{(\gamma-\beta)^2} \cdot \sigma_s^2$$

$$\mathbf{E}\left(p_t^2\right) = \frac{1}{(\gamma-\beta)^2} \cdot \sigma_d^2 + \frac{1}{(\gamma-\beta)^2}\sigma_s^2$$

Combining the above:

$$\begin{aligned}
\hat{\beta} &\to \frac{\mathbf{E}\left(p_t q_t\right)}{\mathbf{E}\left(p_t^2\right)} \\
&= \frac{\frac{\gamma}{(\gamma-\beta)^2} \cdot \sigma_d^2 + \frac{\beta}{(\gamma-\beta)^2} \cdot \sigma_s^2}{\frac{1}{\gamma-\beta} \cdot \sigma_d^2 + \frac{1}{\gamma-\beta}\sigma_s^2} \\
&= \frac{\gamma\sigma_d^2 + \beta\sigma_s^2}{\sigma_d^2 + \sigma_s^2}
\end{aligned}$$

Estimated coefficient: intuition

$$\hat{\beta} \to \frac{\mathbf{E}\left(p_t q_t\right)}{\mathbf{E}\left(p_t^2\right)} = \frac{\gamma\sigma_d^2 + \beta\sigma_s^2}{\sigma_d^2 + \sigma_s^2}$$

Consider the implications of the above limit:

- If $\sigma_s^2 \to \infty$, then $\textit{plim}\left(\hat{\beta}\right) = \beta$. Graphically, this dynamic implies that large shocks to supply causes the regression line to look like the demand line.
- Similarly, if $\sigma_d^2 \to \infty$, then $\textit{plim}\left(\hat{\beta}\right) = \gamma$.

Endogeneity

A regressor is **endogenous** if it is not predetermined

- Math terms: not contemporaneously orthogonal to the error term
- If regressor is correlated with the error term

When a quantity is regressed on a constant and price, it does not estimate the demand nor the supply curve because **price** is **endogenous** (correlated with the error term) in both the demand and supply equations.

- Regressor and error term are related to each other through a **system of equations**
- Endogeneity follows from **simultaneous equation bias**

Neither the demand nor the supply curve is consistently estimated in the above regression because we cannot infer from the data whether the change in price and quantity is due to a demand or a supply shift.

- Suggests that it might be possible to estimate the demand curve if some of the factors that shift the supply curve are observable. If we only were observing shocks to the supply, demand elasticity can be estimated.

Endogeneity problem

Simultaneous Equation Bias

Instrumental variables

IV estimation

IV example

Demand and supply curve **system of equations**:

$$\begin{aligned} \textbf{Demand:} \quad q_t &= \beta p_t + \epsilon_t^d, \\ \textbf{Supply:} \quad q_t &= \gamma p_t + \epsilon_t^s \end{aligned}$$

Consider the variable w_t, the number of freezing days in Florida in year t:

- w_t correlated with price p_t
- w_t is **NOT** correlated with ϵ_t^d

Can use w_t to divide ϵ_t^s into an observable factor (w_t) that is correlated with p_t, and an unobservable factor (ζ_t) that is uncorrelated with w_t and ϵ_t^d:

$$q_t = \gamma_1 p_t + \gamma_2 w_t + \zeta_t.$$

Supply side shift is uncorrelated with the error term in the demand equation (i.e. it is **predetermined**):

- Including w_t is meant to extract from price movements a component that is related to weather (the observed supply shifter), but is uncorrelated with demand shift
- This decomposition can be used to estimate the demand curve by examining the relationship between quantity demanded and the exogenous component of price.

A **predetermined** variable that is correlated with the endogenous regressor is called an **instrumental variable**, or an **instrument**.

Choosing instruments

Instruments must be **relevant**

$$\mathbf{E}\left(w_t \epsilon_t^s\right) \neq 0$$

- Shocks in weather impact supply

Instruments must be **valid**

$$\mathbf{E}\left(w_t \epsilon_t^d\right) = 0$$

- Shocks in weather don't have any impact on demand

Endogeneity problem

Simultaneous Equation Bias

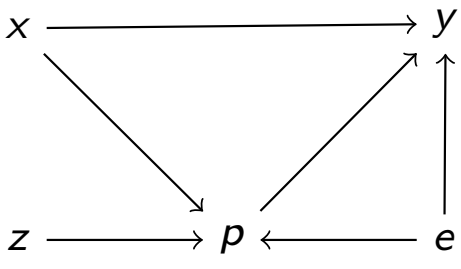
Instrumental variables

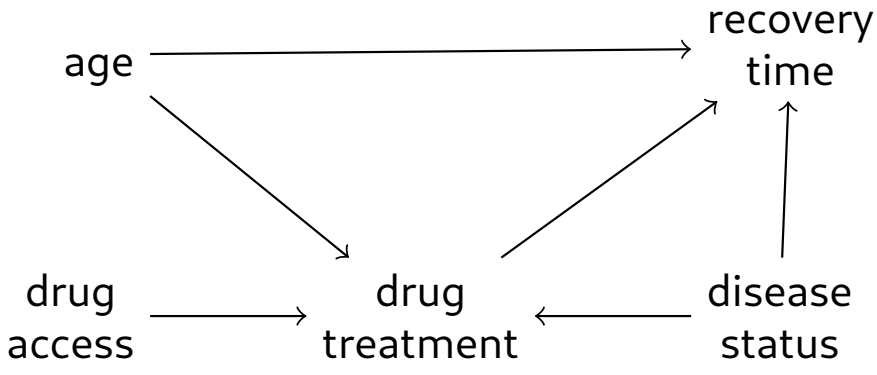
IV estimation

Intent-to-treat set-up

Set-up:

- y: response of interest
- p: treatment variable
- x: observable covariates that directly influence y
- z: instrument; only affects response through influence on treatment
- e: unobserved factors that influence treatment and response





Key takeaways from model:

- Instrument z only impacts outcome y through treatment p
- Instrument z is completely independent of unobserved error e

Two-stage least squares (2SLS) set-up

Set-up (ignore covariates x for now):

- y: response of interest
- p: treatment variable
- e: unobserved factors that influence treatment and response
- z: instrument; only affects response through influence on treatment
  - Relevance: z impacts y only through p
  - Valid: z is independent of e: $\mathbf{E}[e|z] = \mathbf{E}[e] = 0$

Given structural model:

$$y = g(p) + e, \quad \text{Cov}(e, p) \neq 0$$

Can write expectation of structural model **conditional on instrument**:

$$\begin{aligned}\mathbf{E}[y|z] =& \mathbf{E}[g(p) + e|z] \\ =& \mathbf{E}[g(p)|z] + \mathbf{E}[e|z] \\ =& \mathbf{E}[g(p)|z]\end{aligned}$$

$\implies$ given z, y has conditional distribution with mean equal to average of $g(p)$ given z

2SLS motivation

Given structural model:

$$y = g(p) + e, \quad \text{Cov}(e, p) \neq 0$$

Suppose we have a linear treatment model, where $g(p) = \gamma p$

$$y = \gamma p + e, \quad \text{Cov}(e, p) \neq 0$$

Expectation of structural model **conditional on instrument**:

$$\begin{aligned} \mathbf{E}[y|z] =& \mathbf{E}[\gamma p + e|z] \\ =& \gamma \mathbf{E}[p|z] \end{aligned}$$

$\implies$ Can estimate γ with a two-step algorithm (adding covariates x to the conditioning set)

2SLS Algorithm

1. First-stage regression: Regress treatment on instrument (and covariates) using OLS:

$$\mathbf{E}[p|x, z] = \alpha_p + z_i\tau + \mathbf{x}_i'\boldsymbol{\beta}_p$$

Save predicted policy:

$$\hat{p}_i = \hat{\alpha}_p + z_i\hat{\tau} + \mathbf{x}_i'\hat{\boldsymbol{\beta}}_p$$

2. Second-stage regression: Regress response y on predicted policy $\hat{p}_i$ and covariates:

$$\mathbf{E}[y|\hat{p}_i, \mathbf{x}_i] = \alpha_y + \hat{p}_i\gamma + \mathbf{x}_i'\boldsymbol{\beta}_y$$

Can interpret $\hat{\gamma}$ as causal effect of p on y

Intuition:

- First stage create version of p without omitted variable bias
- Second stage uses this modified p to estimate causal effects

Note: standard errors need to be adjusted account for using $\hat{p}$ inputs (technically need to adjust "sandwich" estimator). Use a pre-packaged IV program

IV by hand (cont.)

Linear treatment model with instrument z:

$$y = \gamma p + e, \quad \text{Cov}(e, p) \neq 0$$

Write in terms of covariance:

$$\begin{aligned} \text{Cov}(y, z) &= \text{Cov}(\gamma p + e, z) \\ &= \gamma \text{Cov}(p, z) + \text{Cov}(e, p) \\ &= \gamma \text{Cov}(p, z) \end{aligned}$$

Because instrument z is independent of error e. Dividing each side by $\text{Var}(z)$:

$$\begin{aligned} \frac{\text{Cov}(y, z)}{\text{Var}(z)} &= \gamma \frac{\text{Cov}(p, z}{\text{Var}(z)} \\ \implies \gamma &= \frac{\frac{\text{Cov}(y,z)}{\text{Var}(z)}}{\frac{\text{Cov}(p,z)}{\text{Var}(z)}} = \frac{\text{Reduced form}}{\text{First stage}} \end{aligned}$$

IV by hand

$$\gamma = \frac{\frac{\text{Cov}(y,z)}{\text{Var}(z)}}{\frac{\text{Cov}(p,z)}{\text{Var}(z)}} = \frac{\text{Reduced form}}{\text{First stage}}$$

Numerator (reduced form): regression of y on z

- "Impact" of instrument y on outcome y
- Only happens via p

Denominator (first stage): regression of p on z

- Captures impact of instrument z on treatment p

IV as partial derivatives

$$\gamma = \frac{\frac{\text{Cov}(y,z)}{\text{Var}(z)}}{\frac{\text{Cov}(p,z)}{\text{Var}(z)}} = \frac{\text{Reduced form}}{\text{First stage}}$$

Consider the partial derivatives:

$$\gamma = \frac{\frac{\partial y}{\partial z}}{\frac{\partial p}{\partial z}} = \frac{\partial y}{\partial z} \cdot \frac{\partial z}{\partial p} = \frac{\partial y}{\partial p}$$

What this is showing to us is more subtle than most people appreciate. It is also cooler than most people appreciate.

- Find the impact of instrument z on outcome y
- Find the effect of instrument z on treatment p
- Scale the effect of z on y by the effect of z on p

IV as wald estimator

If instrument z is binary, you can estimate effects using the **Wald estimator**

$$\gamma = \frac{\mathbf{E}[y|z=1] - \mathbf{E}[y|z=0]}{\mathbf{E}[p|z=1] - \mathbf{E}[p|z=0]}$$

Intuition:

- Measure the effect of z on y
- By definition, z only impacts y through p
- Scale this effect by the effect of z on p

Choosing instruments

Instruments need to be reasonable. Example: using weather in the north Atlantic to estimate fish prices in NYC

- Need to confident that weather in the north Atlantic does not impact peoples' willingness to consume fish in New York

There was an explosion in IV papers over the 2000s-2010s, some of which used somewhat unbelievable instruments

- IV is very popular in tech, where randomization can nudge users to treatment, which they may or may not adopt
- In general, IV is most believable when some degree of randomization is employed (i.e. when young men were drafted to the Vietnam lottery)

Weak instruments

When using IV, you are indirectly estimating ATE.

- If your instrument is not directly randomized, you need to choose carefully

Weak instruments:

- An instrument is weak if the instrument z is weakly correlated with the treatment p
- Helpful to look through a computational example to understand this